

Traffictelligence: Advanced Traffic Volume Estimation with Machine Learning

1. Introduction

Project Title:

Traffictelligence: Advanced Traffic Volume Estimation with Machine Learning

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2. Project Overview

Purpose:

The goal of this project is to build an intelligent traffic volume estimation system using machine learning models integrated with a web interface for visualization and interaction. It helps city planners and traffic management departments to predict traffic volume accurately.

Features:

- User authentication and dashboard
 - Real-time data upload and preprocessing
 - Prediction using ML models (Linear, Random Forest, SVR, XGBoost)
 - Visualization of results and comparisons
 - Admin panel for model management
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3. Architecture

Frontend:

Developed using React.js. It provides users with a responsive UI for login, uploading datasets, viewing model predictions, and data visualizations.

Backend:

Built using Node.js and Express.js. It handles API routes, user authentication, ML model requests, and connects with the MongoDB database.

Database:

MongoDB stores user information, uploaded datasets, prediction results, and model logs.

4. Setup Instructions

Prerequisites:

- Node.js (v16 or later)
- MongoDB
- npm or yarn
- Python (for ML model execution)

Installation:

Clone the repository

git clone <https://github.com/your-repo/traffictelligence>

5. Folder Structure

Client (Frontend - React):

client/

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|— src/

| |— components/

| |— pages/

| |— App.js

| |— index.js

Server (Backend - Node.js):

server/

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|— routes/

|— controllers/

|— models/

|— middleware/

|— server.js

6. Running the Application

Frontend:

cd client

npm start

Backend:

cd server

npm start

7. API Documentation

Endpoint	Method	Description
/api/register	POST	Register a new user
/api/login	POST	Authenticate user and return token
/api/upload	POST	Upload dataset for prediction
/api/predict	POST	Trigger ML model prediction
/api/results	GET	Fetch prediction results

8. Authentication

Authentication is handled using **JWT (JSON Web Tokens)**.

- On successful login, a token is issued and stored on the client side.
 - Protected routes require this token for access.
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9. User Interface

The user interface includes:

- Login/Register screen
- Dataset upload page
- Prediction dashboard with model selection
- Graphs and tables showing predicted traffic volumes

(Screenshots can be added manually in the Word document)

10. Testing

Testing Strategy:

- Unit Testing with Jest (backend logic)
- Integration Testing using Postman

- Manual UI Testing for frontend components
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11. Screenshots or Demo

A link to the demo and screenshots can be included:

- GitHub: [Dhanalakshmi816/Trafficelligence-Avanced-Traffic-Volume-Estimation-With-Machine-Learning](https://github.com/Dhanalakshmi816/Trafficelligence-Avanced-Traffic-Volume-Estimation-With-Machine-Learning): TrafficTelligence is an open-source machine learning framework designed to estimate and forecast traffic volume with high precision. It draws from numerous public forks emphasizing the same core functionality.
 - Demo link:
<https://drive.google.com/file/d/1BYF0zRuP1Ja4NbuUHDVwTKCbRZHxFQdT/view?usp=sharing>
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12. Known Issues

- Dataset file format is restricted to CSV only.
 - Model training is done offline – not yet deployed to run dynamically online.
 - Minor UI glitches on smaller screens.
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13. Future Enhancements

- Integrate dynamic training of ML models via backend
- Add real-time data fetching from IoT sensors or APIs
- Enhance UI for mobile devices
- Provide analytics export in PDF or Excel